

The City College of New York

Date: December 9, 2025

Editor-in-Chief

Prof. Amir A. Amini, Ph.D.

IEEE Transactions on Biomedical Engineering (TBME)

Special Issue: “EMBC 2025 Top Papers”

Re: Submission to TBME Special Issue — *Transcranial Focused Ultrasound Stimulation of the Subgenual Anterior Cingulate Reconfigures Network-Level Functional Connectivity: A Sham-Controlled fMRI Study in Healthy Adults*

Dear Prof. Amini,

We are pleased to submit our manuscript entitled *Transcranial Focused Ultrasound Stimulation of the Subgenual Anterior Cingulate Reconfigures Network-Level Functional Connectivity: A Sham-Controlled fMRI Study in Healthy Adults* for consideration in the TBME Special Issue “EMBC 2025 Top Papers.”

This manuscript is a substantially expanded version of our EMBC 2025 paper that was selected among the top 7% of submissions. In accordance with the call, the present article includes **>70% new material**, with extended methods, new analyses, and additional results. Briefly, in a randomized, counterbalanced, sham-controlled study in healthy adults (N=16), we delivered brief tFUS to the subgenual anterior cingulate cortex (sgACC) during resting-state fMRI and quantified whole-brain and network-level functional connectivity (FC) using a preregistered linear mixed-effects framework. We report: (1) a significant time $\times$ condition interaction indicating that sgACC-whole-brain FC increases from pre to post after active tFUS relative to sham; (2) baseline-controlled/ANCOVA and regression-to-the-mean diagnostics demonstrating the result is not explained by spontaneous baseline differences; and (3) network-specificity, wherein sham sessions show a gradual increase in sgACC-DMN coupling, whereas active tFUS preferentially increases sgACC coupling to non-DMN systems, most prominently the cognitive control network.

Compliance and ethical statements. All procedures were approved by the Institutional Review Board of the City University of New York and conducted in accordance with the Declaration of Helsinki. All participants provided written informed consent. The work is original, not under consideration elsewhere, and all authors approve the submission.

Conflicts of interest: None.

We believe this work will be of strong interest to TBME’s readership at the intersection of non-invasive neuromodulation, systems neuroscience, and biomedical imaging, and we appreciate your consideration.

Sincerely,

A handwritten signature in black ink, consisting of a stylized 'J' followed by 'P' and a long horizontal line ending in a series of dots.

Jacek P. Dmochowski, Ph.D. (Corresponding Author)

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